Chapter 6: Mixed Strategies

ECON 5001
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Based on: *Game Theory* by Giacomo Bonanno

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Where We Left Off

Chapter 2: some games have **no** Nash equilibrium

- Matching Pennies, penalty kicks, play-calling...
- Being predictable gets you exploited

Chapter 5: **expected utility** ranks lotteries

- $u(o)$: outcome utility $U(L)$: expected utility

Our goal:

1. Let players **randomize**
2. Use expected utility to evaluate random strategies
3. Find equilibria that always exist

MobLab: “Matrix: Instructor Specified”

- MobLab time!
- You’ll be paired with someone. Row player or Column player
- Two rounds, new partner each round. Each round played once

Round 1

Row	Column: Left	Column: Right
Top	80, 40	40, 80
Bottom	40, 80	80, 40

Round 2

Row	Column: Left	Column: Right
Top	320, 40	40, 80
Bottom	40, 80	80, 40

Feel free to talk! And play it the way you’d actually play it.

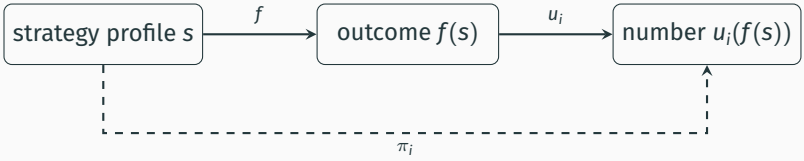
MobLab Results

	% of Rows playing Top	% of Columns playing Left
Round 1		
Round 2		

Strategic-Form Games, Revisited

Strategic-Form Games, Again

Same pipeline as Chapter 2:



One change: u_i is now an **outcome utility** (Chapter 5)

- Numbers are **cardinal**: only linear rescaling allowed
- Bonanno: “games with cardinal payoffs”

Why do we need that?

Example: Two-Point Conversion

Ohio State vs. Michigan. Buckeyes score a TD, go for two

- Ohio State calls: **Run** or **Pass**
- Michigan's defense calls: **Stop Run** or **Stop Pass**
- Simultaneous: neither sees the other's call

Even after both calls, the play can work or fail

Outcome: a **lottery** over {Score, Fail}

Probabilities are made up, for illustration.

Sometimes the Outcome Is a Lottery

	Stop Run	Stop Pass
Run	$\begin{pmatrix} \text{Score} & \text{Fail} \\ 0.4 & 0.6 \end{pmatrix}$	$\begin{pmatrix} \text{Score} & \text{Fail} \\ 0.8 & 0.2 \end{pmatrix}$
Pass	$\begin{pmatrix} \text{Score} & \text{Fail} \\ 0.9 & 0.1 \end{pmatrix}$	$\begin{pmatrix} \text{Score} & \text{Fail} \\ 0.3 & 0.7 \end{pmatrix}$

What number goes in each cell?

Sometimes the Outcome Is a Lottery

	Stop Run	Stop Pass
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What number goes in each cell? Its **expected utility**!

Sometimes the Outcome Is a Lottery

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What number goes in each cell? Its **expected utility**!

Normalize: $u_{OSU}(\text{Score}) = 1$, $u_{OSU}(\text{Fail}) = 0$. Michigan: the reverse

$$U_{OSU}\begin{pmatrix} \text{Score} & \text{Fail} \\ 0.4 & 0.6 \end{pmatrix} = \underline{\qquad\qquad\qquad\qquad} = \Pr(\text{Score})$$

The Reduced Game

Multiply by 100 (linear rescaling). Payoff = % chance of each team's goal

Ohio State \ Michigan	Stop Run	Stop Pass
Run	40, 60	80, 20
Pass	90, 10	30, 70

Rule: Lottery in a cell? Use its expected utility as the payoff number

That's it! From here on, every cell is just a number

Bonanno §6.1 builds this formally (an outcome function that returns lotteries). We'll skip it.

Mixed Strategies

Any Nash Equilibrium?

Ohio State \ Michigan	Stop Run	Stop Pass
Run	40, 60	80, 20
Pass	90, 10	30, 70

Any Nash Equilibrium?

Ohio State \ Michigan	Stop Run	Stop Pass
Run	40, 60	80, 20
Pass	90, 10	30, 70

No cell has both underlined. No (pure) Nash equilibrium!

Any Nash Equilibrium?

Ohio State \ Michigan	Stop Run	Stop Pass
Run	40, <u>60</u>	<u>80</u>, 20
Pass	<u>90</u>, 10	30, <u>70</u>

No cell has both underlined. No (pure) Nash equilibrium!

What would *you* do as the Buckeyes' coach?

Idea: Be unpredictable. **Randomize**!

Definition: Mixed Strategy

A **mixed strategy** σ_i is a probability distribution over player i's strategies

- A lottery over your own strategies!

Ex: $\sigma_{OSU} = \begin{pmatrix} \text{Run} & \text{Pass} \\ p & 1-p \end{pmatrix}$ $\quad$ $\sigma_{UM} = \begin{pmatrix} \text{Stop Run} & \text{Stop Pass} \\ q & 1-q \end{pmatrix}$

- $\sigma_i(s_i)$: probability that σ_i puts on s_i
- Σ_i: set of all mixed strategies of player i
- The old strategies $s_i \in S_i$ are now called **pure strategies**
  - Pure strategy $=$ mixed strategy with probability 1 on one s_i

Assume: players randomize **independently**

Expected Payoff of a Mixed Profile

A mixed profile $\sigma = (\sigma_1, \sigma_2)$ is a lottery over the **cells**

Probability of cell $s = (s_1, s_2)$: $\sigma(s) = \sigma_1(s_1) \times \sigma_2(s_2)$ (independence)

Expected payoff: weight each cell's payoff by the chance of landing in it

$$\underbrace{\Pi_i(\sigma)}_{\text{expected payoff}} = \sum_{s \in S} \underbrace{\sigma(s)}_{\text{prob. of cell } s} \cdot \underbrace{\pi_i(s)}_{\text{payoff in cell } s}$$

Capital Π: payoff from a mixed profile. Small π: from a pure profile

Same idea as *u* vs. *U* in Chapter 5. Bonanno's notation this time.

Example

Ohio State \ Michigan	Stop Run	Stop Pass
Run	40, 60	80, 20
Pass	90, 10	30, 70

OSU: $\begin{pmatrix} \text{Run} & \text{Pass} \\ 0.5 & 0.5 \end{pmatrix}$

Michigan: $\begin{pmatrix} \text{Stop Run} & \text{Stop Pass} \\ 0.8 & 0.2 \end{pmatrix}$

Cell	(Run, Stop Run)	(Run, Stop Pass)	(Pass, Stop Run)	(Pass, Stop Pass)
Probability				

$\Pi_{OSU}(\sigma) =$

$\Pi_{UM}(\sigma) =$

Why We Needed Chapter 5

A mixed profile turns every game into a lottery over cells

Ranking lotteries needs **expected utility**

- Ordinal payoffs only rank the cells
- They can't say if a 50/50 shot at 4 or 1 beats 3 for sure

Thus: Mixed strategies need **cardinal** payoffs

The real payoff from Chapter 5:
Mixed strategies, and **best replies** to mixed strategies

Best Reply to a Mixed Strategy

Michigan plays $\begin{pmatrix} \text{Stop Run} & \text{Stop Pass} \\ q & 1-q \end{pmatrix}$. What should OSU do?

Ohio State \ Michigan	Stop Run	Stop Pass
Run	40, 60	80, 20
Pass	90, 10	30, 70

$\Pi_{OSU}(\text{Run},\ q) =$

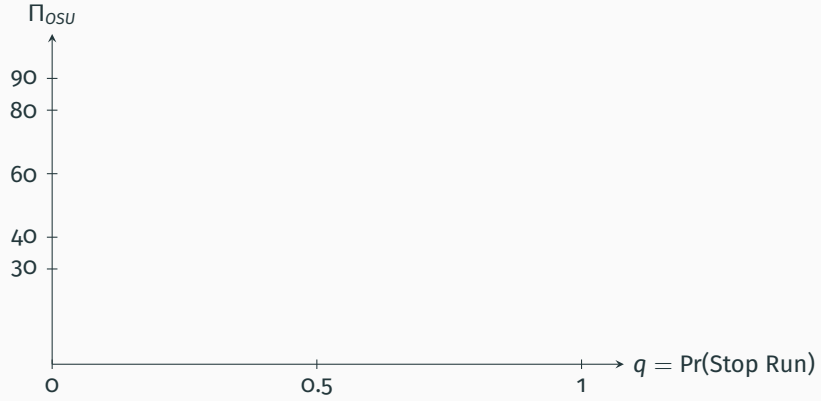
$\Pi_{OSU}(\text{Pass},\ q) =$

If $q = 0.8$: Run gives _____, Pass gives _____

OSU's best reply: _____

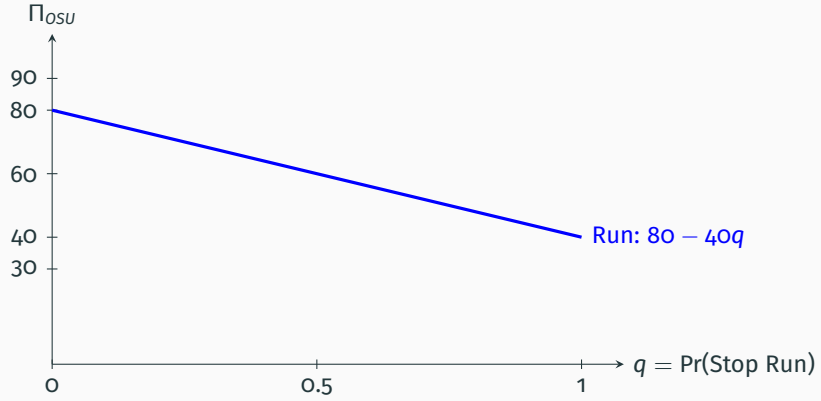
Michigan's mix is a lottery, so each pure strategy of OSU's gets an expected payoff.

Seeing It in a Graph



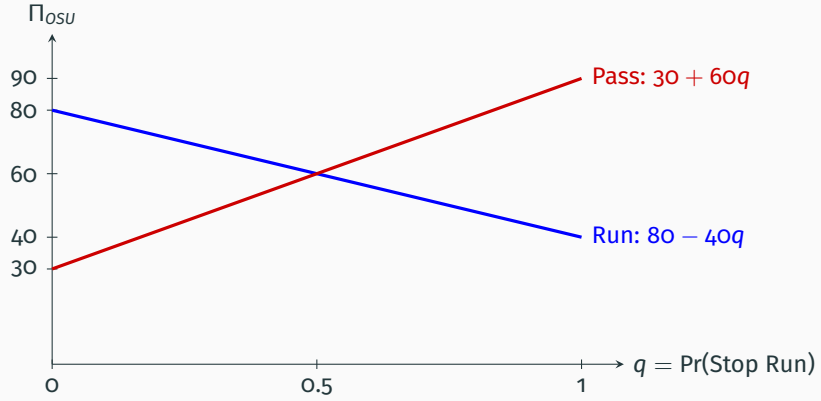
Plot each pure strategy's expected payoff against q

Seeing It in a Graph



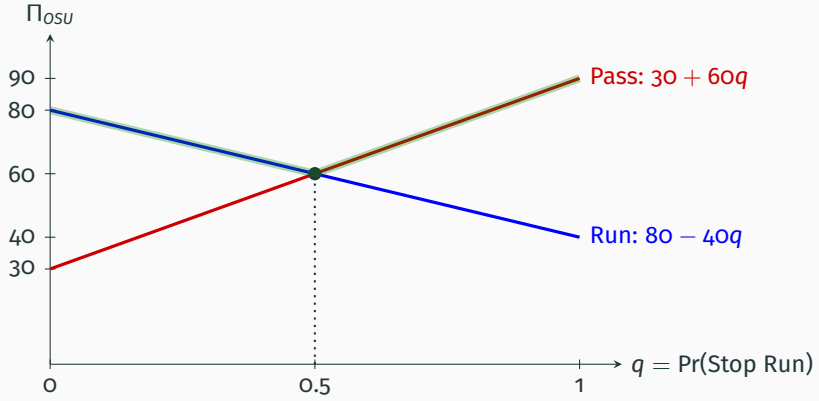
Plot each pure strategy's expected payoff against q

Seeing It in a Graph



Plot each pure strategy's expected payoff against q

Seeing It in a Graph



Best reply = the higher line. They cross at $q =$ _____

OSU's Best Reply

$$BR_{OSU}(q) = \begin{cases} \text{Run} & \text{if } q < \frac{1}{2} \\ \text{_____} & \text{if } q = \frac{1}{2} \\ \text{Pass} & \text{if } q > \frac{1}{2} \end{cases}$$

Fact: A mix of Run and Pass is a best reply **only** when Run and Pass tie

- Otherwise, shift weight to the better one!

Fact: When they tie, **every** mix is a best reply

In general: σ_i is a best reply $\iff$ every pure strategy it uses is a best reply

Definition: Mixed-Strategy Nash Equilibrium

In words: every player's mix is a best reply to the other players' mixes

A mixed profile $\sigma^* = (\sigma_1^*, \ldots, \sigma_n^*)$ is a **Nash equilibrium** if, for every player i,

$$\underbrace{\Pi_i(\sigma_i^*, \sigma_{-i}^*)}_{\text{payoff from equilibrium mix}} \geq \underbrace{\Pi_i(\sigma_i, \sigma_{-i}^*)}_{\text{payoff from any other mix}} \qquad \text{for all } \sigma_i \in \Sigma_i$$

Same as Chapter 2: $\sigma_i^* \in BR_i(\sigma_{-i}^*)$ for every i

Fact: Enough to check **pure** deviations $s_i \in S_i$

A pure-strategy NE is still a NE when mixing is allowed (Bonanno p. 211).

Nash's Theorem

Theorem (Nash, 1951)
Every finite game has **at least one** Nash equilibrium in mixed strategies.

Finite: finitely many players, each with finitely many strategies.

- Matching Pennies: no pure NE, but a mixed one
- Two-point conversion: no pure NE, but a mixed one
- This was the big result in his 26-page thesis

Proof uses a fixed-point theorem. Not covered here

Existence needs **mixing**, and mixing needs **expected utility**

Finding Mixed Equilibria

The Indifference Principle

In words: if you mix between two strategies in equilibrium, they must pay the same

If σ_i^* puts positive probability on both s_i and s_i', then

$$\underbrace{\Pi_i(s_i, \sigma_{-i}^*)}_{\text{payoff from } s_i} = \underbrace{\Pi_i(s_i', \sigma_{-i}^*)}_{\text{payoff from } s_i'} = \underbrace{\Pi_i(\sigma^*)}_{\text{equilibrium payoff}}$$

Why?

Lesson: My mix is set to make **you** indifferent. Yours makes **me** indifferent

Bonanno Theorem 6.3.1

Recipe: Mixed NE in a 2×2

1. Run IDSDS first. Dominated strategies get probability 0
2. Underline. Note any pure NE
3. P2 plays q on their first strategy. Find the q that makes **P1** indifferent
4. P1 plays p on their first strategy. Find the p that makes **P2** indifferent
5. Both strictly between 0 and 1? That's a mixed NE

Bigger games: also check that unused strategies don't do better

Solving the Two-Point Game

Ohio State \ Michigan	Stop Run	Stop Pass
Run	40, 60	80, 20
Pass	90, 10	30, 70

Step 1: Michigan's q makes OSU indifferent

$q^* =$

Solving the Two-Point Game

Ohio State \ Michigan	Stop Run	Stop Pass
Run	40, 60	80, 20
Pass	90, 10	30, 70

Step 2: OSU's p makes Michigan indifferent

$p^* =$

Mixed NE:

OSU scores _____% of the time

Back to MobLab

Round 1

Row \ Column	Left	Right
Top	80, 40	40, 80
Bottom	40, 80	80, 40

Round 2

Row \ Column	Left	Right
Top	320, 40	40, 80
Bottom	40, 80	80, 40

Round 1 mixed NE:

Round 2 mixed NE:

Only Row's payoff changed. Whose mix changes in the NE??

Back to MobLab

	% Rows playing Top		% Columns playing Left	
	Round 1	Round 2	Round 1	Round 2
Nash prediction	50	50	50	12.5
Goeree & Holt (2001)	$\approx$ 50	96	$\approx$ 50	16
Our class				

Lab result: Rows chase the 320. NE says Row's own payoffs shouldn't matter! Columns anticipate it and play Left even less

Not in the book. Source: Goeree & Holt (2001), "Ten Little Treasures of Game Theory and Ten Intuitive Contradictions," *American Economic Review* 91(5).

Pure and Mixed Together

Remember Sunday Morning?

Alex and Blair, dead phones: Buckeye Donuts or HangOverEasy?

		Blair: BuckyDs	Blair: HangOE
Alex	BuckyDs	4, 3	2, 1
Alex	HangOE	1, 2	3, 4

Now read the payoffs as **cardinal** (outcome utilities)

Remember Sunday Morning?

Alex and Blair, dead phones: Buckeye Donuts or HangOverEasy?

		Blair: BuckyDs	Blair: HangOE
Alex	BuckyDs	4, 3	2, 1
Alex	HangOE	1, 2	3, 4

Now read the payoffs as **cardinal** (outcome utilities)

Pure NE: _____

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Alex	BuckyDs	4, 3	2, 1
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Now read the payoffs as **cardinal** (outcome utilities)

Pure NE: _____

Is there also a mixed NE?

Sunday Morning: The Mixed NE

		Blair: BuckyDs	Blair: HangOE
Alex	BuckyDs	4, 3	2, 1
Alex	HangOE	1, 2	3, 4

Blair's q makes Alex indifferent:

Alex's p makes Blair indifferent:

Payoffs at the mixed NE: Alex _____ Blair _____ Better or worse than the pure NE?

What if Alex's 4 were a 10? Same ranking. Whose mix changes?

Indifference Isn't Enough

Player 1 \ Player 2	D	E
A	3, 0	0, 2
B	0, 2	3, 0
C	2, 0	2, 1

Try: P1 plays $\begin{pmatrix} A & B & C \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}$, P2 plays $\begin{pmatrix} D & E \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$

P1 indifferent between A and B? $\Pi_1(A) =$ $\Pi_1(B) =$

Indifference Isn't Enough

Player 1 \ Player 2	D	E
A	3, 0	0, 2
B	0, 2	3, 0
C	2, 0	2, 1

Try: P1 plays $\begin{pmatrix} A & B & C \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}$, P2 plays $\begin{pmatrix} D & E \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$

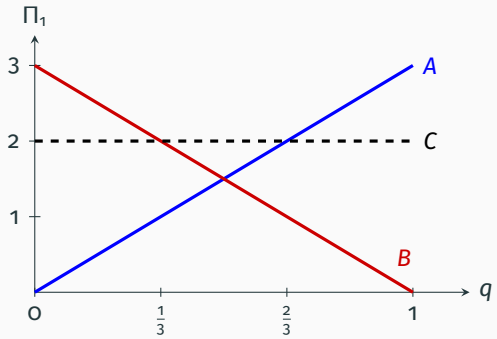
P1 indifferent between A and B? $\Pi_1(A) =$ $\Pi_1(B) =$

But $\Pi_1(C) =$ _____ So it's **not** a NE!

Lesson: Indifference is necessary, not sufficient. Check the unused strategies!

Best Replies With Three Strategies

Player 1 \ Player 2	D	E
A	3, 0	0, 2
B	0, 2	3, 0
C	2, 0	2, 1



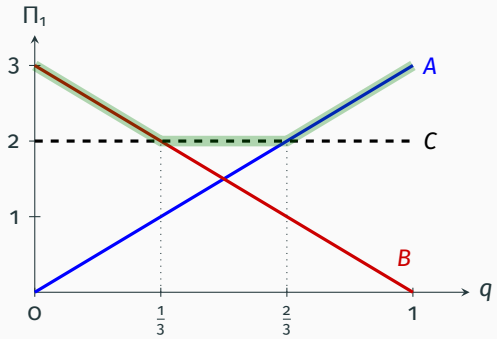
$q = \Pr(D)$

$\Pi_1(A, q) = 3q$
$\Pi_1(B, q) = 3 - 3q$
$\Pi_1(C, q) = 2$

$BR_1(q)$:

Best Replies With Three Strategies

Player 1 \ Player 2	D	E
A	3, 0	0, 2
B	0, 2	3, 0
C	2, 0	2, 1



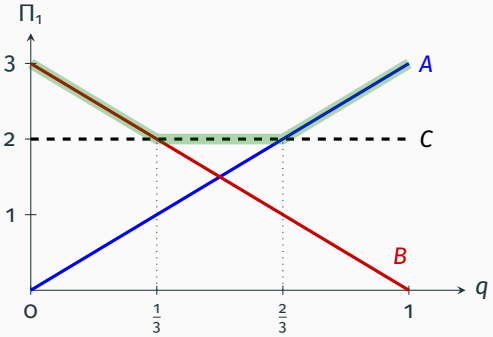
$q = \Pr(D)$

$\Pi_1(A, q) = 3q$
$\Pi_1(B, q) = 3 - 3q$
$\Pi_1(C, q) = 2$

$BR_1(q)$: B if $q < \frac{1}{3}$ C if $\frac{1}{3} < q < \frac{2}{3}$ A if $q > \frac{2}{3}$

Best Replies With Three Strategies

Player 1 \ Player 2	D	E
A	3, 0	0, 2
B	0, 2	3, 0
C	2, 0	2, 1



$q = \Pr(D)$

$\Pi_1(A, q) = 3q$
$\Pi_1(B, q) = 3 - 3q$
$\Pi_1(C, q) = 2$

$BR_1(q)$: B if $q < \frac{1}{3}$ C if $\frac{1}{3} < q < \frac{2}{3}$ A if $q > \frac{2}{3}$

So the NE must mix B & C (at $q = \frac{1}{3}$), or A & C (at $q = \frac{2}{3}$). Which works?

What Does Mixing Mean?

At a mixed NE you're **indifferent**. So why mix exactly 60/40???

Three readings:

1. **Literal randomizing.** Coaches, kickers, poker players
2. **Beliefs.** σ_2 is what P1 *believes* P2 will do
   - P2 may not randomize at all. P1 just can't predict them
3. **Populations.** 60% of coaches run, 40% pass

Bonanno calls mixed NE "conceptually problematic." Agree?

Mixing in the Real World

Lots of situations boil down to “be unpredictable”:

- **Penalty kicks.** Kick left or right? Dive left or right?
- **Tennis serves.** To the forehand or the backhand?
- **Play-calling, pitch selection.** Run/pass. Fastball/curveball
- **Poker.** Bluff with a bad hand, or fold?
- **Inspections.** Tax audits, parking tickets, drug tests, fare checks
- **Security.** LAX randomizes its checkpoints and patrols

Lab/field result: Pro soccer and tennis players mix close to NE rates

Not in the book. Sources: Walker & Wooders (2001, *AER*); Chiappori, Levitt & Groseclose (2002, *AER*); Palacios-Huerta (2003, *REStud*); Pita et al. (2008, AAMAS).

But Those Are More Complicated

Real versions add:

- More than two strategies (left, middle, right)
- Repeated play: patterns can be spotted and exploited
- Players differ in skill (a left-footed kicker)
- Payoffs nobody knows exactly

Core idea survives: if you're predictable, you get exploited

Dominance, Revisited

Remember This One?

From Chapter 2: M is never a best reply to a **pure** strategy, but isn't dominated

Player 1 \ Player 2	L	C	R
U	4, 1	2, 0	4, 2
M	3, 2	3, 1	3, 0
D	0, 0	4, 2	0, 1

Suppose P2 plays $\begin{pmatrix} L & C & R \\ 0.4 & 0.6 & 0 \end{pmatrix}$

$\Pi_1(U) =$ $\Pi_1(M) =$ $\Pi_1(D) =$

Remember This One?

From Chapter 2: M is never a best reply to a **pure** strategy, but isn't dominated

Player 1 \ Player 2	L	C	R
U	4, 1	2, 0	4, 2
M	3, 2	3, 1	3, 0
D	0, 0	4, 2	0, 1

Suppose P2 plays $\begin{pmatrix} L & C & R \\ 0.4 & 0.6 & 0 \end{pmatrix}$

$\Pi_1(U) =$ $\Pi_1(M) =$ $\Pi_1(D) =$

M **is** a best reply, to a **mixed** strategy!

Dominated by a Mix

Player 1 \ Player 2	D	E
A	0, 1	4, 0
B	1, 2	1, 4
C	2, 0	0, 1

Is B dominated by A? By C?

Try the mix $\begin{pmatrix} A & B & C \\ \frac{1}{3} & 0 & \frac{2}{3} \end{pmatrix}$: vs. D: vs. E:

Dominated by a Mix

Player 1 \ Player 2	D	E
A	0, 1	4, 0
B	1, 2	1, 4
C	2, 0	0, 1

Is *B* dominated by *A*? By *C*?

Try the mix $\begin{pmatrix} A & B & C \\ \frac{1}{3} & 0 & \frac{2}{3} \end{pmatrix}$: vs. *D*: vs. *E*:

B is **strictly dominated by a mixed strategy**
So *B* is never a best reply, to anything P2 might do

Pearce's Theorem

Theorem (Pearce, 1984)
In a two-player game, s_i is never a best reply to any mixed strategy of the opponent

$$\iff$$

s_i is strictly dominated by some mixed strategy of player i

- Fixes Chapter 2's gap: "never a best reply" = "dominated by a mix"
- Opponent's mix = my **belief** about them
  - Dominated by a mix $\iff$ irrational under **every** belief

Three or more players: true if beliefs may be correlated (Bonanno §6.4).

Cardinal IDSDS

Same as IDSDS, but also delete strategies dominated by **mixes**

- Survivors are called **rationalizable** strategies

Player 1 \ Player 2	D	E	F
A	3, 4	2, 1	1, 2
B	0, 0	1, 3	4, 1
C	1, 4	1, 4	2, 6

Step 1:

Step 2:

Step 3:

Step 4:

Careful: needs players to know each other's **cardinal** payoffs. Realistic?

Let’s Work One Out

Gameday Parking

Jordan drives to campus for the Michigan game. Three options:

- **Pay** for a gameday lot
- Park illegally in a **West** Campus lot
- Park illegally on **Neil** Ave

CampusParc has one tow crew. It can patrol West Campus or Neil Ave

Jordan \ CampusParc	Patrol West	Patrol Neil
Pay	1, 0	1, 0
West	0, 2	4, 0
Neil	3, 0	0, 1

Jordan picks the row, CampusParc the column. Payoffs: Jordan, CampusParc. Hypothetical numbers.

Gameday Parking

1. Is any strategy strictly dominated by a **pure** strategy?

2. Is any strategy strictly dominated by a **mixed** strategy? Delete it

3. Is there a pure-strategy NE?

4. Find the mixed NE and both players' expected payoffs.
   Check that Jordan really doesn't want to Pay

5. CampusParc now earns 4 (not 2) for a tow at West Campus.
   Whose mix changes? Does Jordan's payoff change?

What Have We Learned, Charlie Brown?

- Lottery in a cell? Its expected utility is the payoff
- **Mixed strategy**: a lottery over your own pure strategies
- $\Pi_i(\sigma)$: weight each cell's payoff by its probability
  - Needs **cardinal** payoffs. That's why we did Chapter 5
- **Mixed NE**: every mix is a best reply to the others
- **Nash (1951)**: every finite game has one
- **Indifference principle**: my mix makes *you* indifferent
  - Necessary, not sufficient. Check unused strategies
- **Pearce**: never a best reply $\iff$ dominated by a mix

Next: mixing in game trees (behavioral strategies)

Reading & Practice

- Bonanno, *Game Theory*, Chapter 6
  - §6.1: skim. Lotteries in cells, done formally
  - §6.2–6.4: read carefully
- Exercises: §6.5.2 (6.3, 6.4), §6.5.3 (6.7, 6.8, 6.10), §6.5.4 (6.12)
  - 6.13 and 6.14 are harder. 6.15 (the dog in the river) is a challenge
- Solutions in §6.6. Try them first